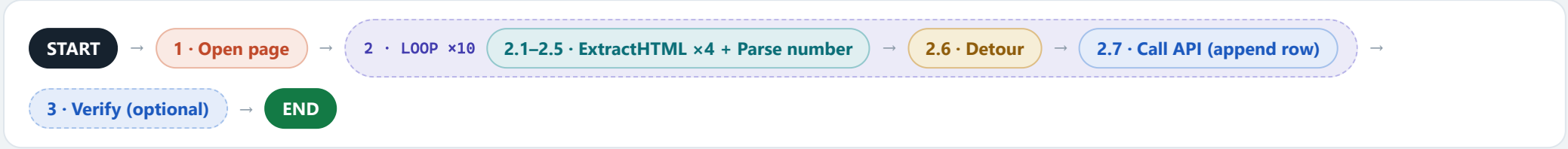


Wikipedia → Google Sheets

An RPA workflow that extracts **Year, Winner, Score, Runners-up** from the first 10 rows of the “*List of FIFA World Cup finals*” table and appends each row to a Google Sheet via the Google Sheets API (configured exactly as a Postman request) — built **exclusively from the 9 steps in the provided IE Step Library**.

Candidate · Bhunesh Bansal **Date** · 8 July 2026 **Source** · en.wikipedia.org/wiki/List_of_FIFA_World_Cup_finals **Target** · Google Sheet — tab Sheet1, columns A:D



A Reading guide — the Step Library

Constraint honored: every executable node is one of the library’s 9 steps — 6 are used, the other 3 are listed with why they’re unnecessary. The brief says append to an “Excel sheet”; the Google Sheets API only writes to a *native* Google Sheet, so the target is a converted Sheet (setup S1).

Open page starts the RPA browsing session used ×1	Loop fixed-count iteration, index from 1 used ×1 (10 iterations)	ExtractHTML DOM path in → element text out used ×4 per iteration
Parse number numeric string → number + row-validity gate used ×1 per iteration	Detour “if” — run branch when true, then resume used ×1 per iteration	Call API configured exactly like Postman used ×1 per iteration (+1 optional QA)
Fill text field types into an input not needed – read-only page, nothing to type	Click simulated mouse click deliberately avoided – clicking a column header would re-sort rows and scramble tr[index]	Check Box toggles a checkbox not needed – page has no checkboxes

B One-time setup (before the workflow runs)

Preconditions only — these are **not** workflow steps and use nothing from the Step Library.

PREPARED ONCE, REUSED ON EVERY RUN

S1 Target spreadsheet
Create a Google Sheet (or convert the uploaded Excel file to Google Sheets format — the API can only write to native Sheets). Tab `Sheet1` ; headers `Year | Winner | Score | Runners-up` in A1:D1. Copy the **spreadsheetId** from the URL between `/d/` and `/edit` .

S2 Google Cloud
Create a project → enable the **Google Sheets API** → create OAuth 2.0 client credentials (the sheet’s owner account must grant access).

S3 Postman auth
OAuth 2.0 (Authorization Code) → obtain `{{ACCESS_TOKEN}}` with scope `.../auth/spreadsheets` . Full config in panel D.

S4 Capture DOM paths
Per the resource video: DevTools → Inspect element → right-click the cell node → **Copy** → **Copy XPath**. Replace the row number with `{{index}}+1` . Paths live-verified in panel E.

C The workflow — flowchart

One straight spine: open the page, loop over the 10 finals rows, and inside every iteration extract 4 fields, type-check the year, guard with a Detour, and append the row via one API call. Loop index `i` does triple duty: iteration counter ↔ table row `tbody/tr[i+1]` (row 1 of tbody is the header) ↔ sheet row `i+1`.



1 Open the source page

Open page

Begins the RPA browsing session (per library note) and loads the article that hosts the target table.

INPUThttps://en.wikipedia.org/wiki/List_of_FIFA_World_Cup_finals

Page URL — the only required input.

OUTPUTLive browser session with the page fully rendered. The table keeps its header row as the first `<tbody>` row — verified in a live browser on 7 Jul 2026 (today's page no longer relocates it into `<thead>` the way the May-2024 resource video shows) — so data rows are `tbody/tr[2] ... tbody/tr[23]`, and the 1930 final is `tr[2]`.

Keep default (chronological) sort. No clicks on column headers — sorting re-orders `tr` indices and would corrupt the extraction. The table renders on page load; no interaction is needed.

page ready

2 Loop — once per finals row

Loop

Input: number of iterations = 10 · Output: index `i` = 1, 2, ... 10 (library: index starts from 1)

`i = 1 → tr[2] = 1930 ... i = 10 → tr[11] = 1974` error inside an iteration ⇒ run continues at next index (library note)

rows are processed top-to-bottom, so sheet order = table order

row selector `tr[{{index}}+1]` stays valid XPath after substitution — `tr[1+1] ≡ tr[2]` (verified live)

how: the Loop substitutes `{{index}}` into each DOM-path string, then ExtractHTML resolves it

2.1 Extract “Year”

ExtractHTML

INPUT

//*[@id="mw-content-text"]/div[2]/section[2]/table[3]/tbody/tr[{{index}}+1]/th/a

Element DOM path (XPath) — year link inside the row-header cell. `+1` skips the header row.

OUTPUTText → `year_text` e.g. `i=1 → "1930"`

2.2 Convert year to a number

Parse number

INPUT

`year_text`

String, e.g. `"1930"`

OUTPUT

`year`

Numeric, e.g. `1930` — appended unquoted so column A holds a real number, not text.

Primary purpose is a row-guard, not the conversion: if markup ever shifts and a non-numeric string reaches this step, it errors → the Loop skips to the next index (library behaviour), so a malformed row can never reach the sheet.

Library type note: the step's *description* says it returns an `Int` while its *Output* column lists “String” — the guard holds either way (see edge case #11).

2.3 Extract “Winner”

ExtractHTML

INPUT

//*[@id="mw-content-text"]/div[2]/section[2]/table[3]/tbody/tr[{{index}}+1]/td[1]

Whole Winners cell. Deliberately **not** the country link: its own path varies row to row (`td[1]/a` on 1930, `td[1]/span/a` on 1954 — verified live), so the cell is the only tail that works for all 10 rows.

OUTPUTText → `winner` e.g. `i=1 → "Uruguay"` (trimmed — see edge case #9)

2.4 Extract “Score”

ExtractHTML

INPUT

//*[@id="mw-content-text"]/div[2]/section[2]/table[3]/tbody/tr[{{index}}+1]/td[2]

Whole cell — keeps the “(a.e.t.)” qualifier where a final went to extra time (1934, 1966).

OUTPUTText → `score` e.g. `i=1 → "4-2"`, `i=2 → "2-1 (a.e.t.)"`

Scores use an **en dash** (`-`, `U+2013`), not a hyphen. Row 4 (1950) carries a footnote marker in the same cell; the library has no string-editing step to strip it, so the whole cell is the honest choice under the constraint — see edge case #3.

2.5 Extract “Runners-up”

ExtractHTML

INPUT

//*[@id="mw-content-text"]/div[2]/section[2]/table[3]/tbody/tr[{{index}}+1]/td[3]

Whole Runners-up cell (country link is span-wrapped on every row — cell-level is uniform).

OUTPUTText → `runners_up` e.g. `i=1 → "Argentina"` (trimmed)

next index · `i ≤ 10` · repeat 0

2.6 ◇ Append guard — is the row complete?

Detour

Library: “conditional execution like an ‘if’ statement; if true, run branch then resume the run.” One branch, one condition.

INPUT

winner ≠ "" AND score ≠ "" AND runners_up ≠ ""

year

 is already validated by step 2.2 — this guards the three text fields against a silent partial extraction.

OUTPUT

true → branch executes (2.7), then the run resumes · false → branch skipped, loop moves on

condition FALSE

Branch does not run. Nothing is appended — **no partial rows ever reach the sheet.**

Run resumes → next index.

condition TRUE → run branch

2.7 Append this row to Google Sheets

Call API

One POST per row, configured exactly as a Postman request — full spec in panel D.

INPUT

POST .../values/Sheet1!A:D:append

Body → "values": [[{{year}}, "{{winner}}", "{{score}}", "{{runners_up}}"]]

e.g. i=1 → [[1930, "Uruguay", "4-2", "Argentina"]]

OUTPUT

HTTP 200 · updates.updatedRange = "Sheet1!A{{index+1}}:D{{index+1}}" (i=1 → A2:D2)

If the call fails (401/429/network), the iteration errors and the Loop **skips to the next index** — one bad row never blocks the other nine.

both paths converge → run resumes → Loop advances to next index

after iteration 10 → exit loop ▼

all rows processed

3 Verify the write (optional QA — recommended)

Call API

INPUT

GET
https://sheets.googleapis.com/v4/spreadsheets/{{SPREADSHEET_ID}}/values/Sheet1!A:D

OUTPUT

200 with values array of **11 rows** (1 header + 10 data) — confirms every append landed.

END

Final state: Sheet1!A2:D11 = the first 10 FIFA World Cup finals, 1930 → 1974 (full data in panel F)

D The API call — Google Sheets values:append via Postman

Exact configuration for step 2.7. Everything in {{double_braces}} is a Postman/workflow variable.

POSThttps://sheets.googleapis.com/v4/spreadsheets/{{SPREADSHEET_ID}}/values/{{RANGE}}:append

PATH VARIABLES

VARIABLE	VALUE / WHERE IT COMES FROM
SPREADSHEET_ID	From the sheet URL: docs.google.com/spreadsheets/d/<this>/edit
RANGE	Sheet1!A:D — the table to append to (Postman URL-encodes ! and : automatically)

QUERY PARAMETERS

PARAM	WHY
valueInputOption=RAW	Write values literally. USER_ENTERED would let Sheets re-interpret strings (dash-separated scores can coerce to dates); RAW is deterministic.
insertDataOption=INSERT_ROWS	Always add new rows below the existing table — never overwrite.
includeValuesInResponse=true	Optional: response echoes what was written — instant per-row verification.

HEADERS

Content-Type	application/json
Authorization	Bearer {{ACCESS_TOKEN}} (set automatically by Postman's OAuth 2.0 helper)

AUTH — POSTMAN OAUTH 2.0 (AUTHORIZATION CODE)

Auth URL	https://accounts.google.com/o/oauth2/v2/auth
Token URL	https://oauth2.googleapis.com/token
Scope	https://www.googleapis.com/auth/spreadsheets
Callback	https://oauth.pstmn.io/v1/callback

REQUEST BODY — TEMPLATE (ITERATION I)

```
{  "range": "Sheet1!A:D",  "majorDimension": "ROWS",  "values": [    [ {{year}}, "{{winner}}", "{{score}}", "{{runners_up}}" ]  ]}
```

CONCRETE EXAMPLE — ITERATION 1

```
{  "range": "Sheet1!A:D",  "majorDimension": "ROWS",  "values": [    [ 1930, "Uruguay", "4-2", "Argentina" ]  ]}
```

SUCCESS RESPONSE — 200 OK (ITERATION 1)

```
{  "spreadsheetId": "{{SPREADSHEET_ID}}",  "tableRange": "Sheet1!A1:D1",  "updates": {    "updatedRange": "Sheet1!A2:D2",    "updatedRows": 1,    "updatedColumns": 4,    "updatedCells": 4  }}
```

FAILURE MODES

401	Token expired → refresh in Postman (Get New Access Token).
403	Sheets API not enabled, or account lacks edit access to the sheet.
404	Wrong SPREADSHEET_ID.
429	Quota (60 write req/min/user). 10 sequential appends sit far below it.

Inside the Loop, any failure only skips that row (library note) — remaining iterations still append. :append is **not idempotent**: re-running the workflow duplicates rows, so clear A2:D11 before a re-run.

E DOM paths — captured & verified

Method from the resource video (which demonstrates the technique on this very table): Inspect the element → right-click the **cell** (th/td) in the Elements panel → **Copy** → **Copy XPath** → replace the row number with `{{index}}+1`. Every path below returned the expected text for **all 10 rows in a live Chrome session** — 40/40 checks passed, and the templated predicate `tr[1+1]` was confirmed to resolve to `tr[2]`.

FIELD	XPATH TEMPLATE (INPUT TO EXTRACTHTML)	OUTPUT @ I = 1
Year	<code>//*[@id="mw-content-text"]/div[2]/section[2]/table[3]/tbody/tr[{{index}}+1]/th/a</code>	"1930"
Winner	<code>//*[@id="mw-content-text"]/div[2]/section[2]/table[3]/tbody/tr[{{index}}+1]/td[1]</code>	"Uruguay " → trim
Score	<code>//*[@id="mw-content-text"]/div[2]/section[2]/table[3]/tbody/tr[{{index}}+1]/td[2]</code>	"4-2"
Runners-up	<code>//*[@id="mw-content-text"]/div[2]/section[2]/table[3]/tbody/tr[{{index}}+1]/td[3]</code>	" Argentina" → trim

Why these differ from the resource video — three verified changes since May 2024. ① **Prefix:** the video copies `//*[@id="mw-content-text"]/div[1]/table[4]/...`; Wikipedia now uses section-wrapped markup, so the same click yields `div[2]/section[2]/table[3]` (the old prefix resolves to nothing — re-copy at build time; the workflow keeps it in one variable). ② **Row offset:** in the video the sorter moves the header row into `<thead>`, making data row `i = tr[i]`; on today's page the header stays as `tbody` row 1, so data row `i = tr[i+1]`. ③ **Anchor nesting:** the video's `td[1]/a` works for 1930 but fails on rows where the link is span-wrapped (`td[1]/span/a`, e.g. 1954) — cell-level paths are the only tails uniform across all 10 rows.

F Expected result — the sheet after the run

Real values from the live table (not placeholders). The first 10 rows span 1930–1974: Wikipedia lists no rows for 1942/1946 (tournaments cancelled, WWII), so a fixed loop bound of 10 is exact.

	A	B	C	D
1	Year	Winner	Score	Runners-up
2	1930	Uruguay	4–2	Argentina
3	1934	Italy	2–1 (a.e.t.)	Czechoslovakia
4	1938	Italy	4–2	Hungary
5	1950	Uruguay	2–1 [n 3]	Brazil
6	1954	West Germany	3–2	Hungary
7	1958	Brazil	5–2	Sweden
8	1962	Brazil	3–1	Czechoslovakia
9	1966	England	4–2 (a.e.t.)	West Germany
10	1970	Brazil	4–1	Italy
11	1974	West Germany	2–1	Netherlands

Row 5’s score keeps Wikipedia’s footnote marker **[n 3]** (the 1950 “final” was the deciding match of a round-robin group) — see edge case #3 for the trade-off and the cleaner alternative.

Execution trace — the run, iteration by iteration

What each loop pass extracts and sends. Every value below was produced by evaluating the panel-E XPaths in a live browser session.

I	SOURCE ROW	REQUEST BODYVALUES (STEP 2.7)	200 →UPDATEDRANGE
1	tbody/tr[2]	[[1930, "Uruguay", "4–2", "Argentina"]]	Sheet1!A2:D2
2	tbody/tr[3]	[[1934, "Italy", "2–1 (a.e.t.)", "Czechoslovakia"]]	Sheet1!A3:D3
3	tbody/tr[4]	[[1938, "Italy", "4–2", "Hungary"]]	Sheet1!A4:D4
4	tbody/tr[5]	[[1950, "Uruguay", "2–1[n 3]", "Brazil"]]	Sheet1!A5:D5
5	tbody/tr[6]	[[1954, "West Germany", "3–2", "Hungary"]]	Sheet1!A6:D6
6	tbody/tr[7]	[[1958, "Brazil", "5–2", "Sweden"]]	Sheet1!A7:D7
7	tbody/tr[8]	[[1962, "Brazil", "3–1", "Czechoslovakia"]]	Sheet1!A8:D8
8	tbody/tr[9]	[[1966, "England", "4–2 (a.e.t.)", "West Germany"]]	Sheet1!A9:D9
9	tbody/tr[10]	[[1970, "Brazil", "4–1", "Italy"]]	Sheet1!A10:D10
10	tbody/tr[11]	[[1974, "West Germany", "2–1", "Netherlands"]]	Sheet1!A11:D11

Detour evaluates **true** on all 10 iterations (every field non-empty), so all 10 branches run — 10 POSTs, 10 × updatedRows: 1, zero skips.

G Edge cases & design decisions

#1 “First 10 rows” ≠ “first 10 World Cups by year”

1942 and 1946 were cancelled and have no table rows, so rows 1–10 cleanly map to 1930–1974. The loop bound of 10 needs no gap-handling.

#3 The 1950 footnote marker

Row 4’s score cell embeds a superscript note → cell text is 2–1[n 3]. Chosen: extract the whole cell (keeps “(a.e.t.)” on rows 2 & 8) and accept the marker. Alternative: point ExtractHTML at .../td[2]/a for a pure 2–1 — at the cost of dropping “(a.e.t.)”.

#5 XPaths drift; the method doesn’t

Three things changed between the resource video (May 2024) and today — all caught live: the table prefix (div[1]/table[4] → div[2]/section[2]/table[3]), the header row’s home (<thead> → first tbody row, hence tr[{{index}}+1]), and the country-link nesting (td[1]/a vs td[1]/span/a , hence cell-level paths). The prefix lives in one variable, so the next drift is a one-field fix.

#7 Why Parse number earns its place

Parse number converts "1930" → 1930. Its decisive value is the per-row validity gate (a non-numeric year ⇒ error ⇒ loop skips that index); the numeric column A is a secondary nicety, subject to the library’s type ambiguity in #11.

#2 Append per row, not one batch call

The library has no accumulator step to build a 10-row array across iterations, and per-row appends pair perfectly with the Loop’s skip-on-error rule: one bad row costs one row, not the whole write. 10 sequential calls sit far under the 60 write-req/min quota.

#4 Scores contain an en dash

4–2 uses U+2013, not a hyphen. With valueInputOption=RAW the string is stored verbatim; no risk of Sheets coercing a dash-separated pair into a date.

#6 Never sort before extracting

The table is user-sortable; clicking a header re-orders <tr> elements and would silently remap every index. The workflow performs zero clicks — extraction happens on the default chronological order.

#8 Re-runs are not idempotent

:append always adds rows; running the workflow twice writes 20 rows. Clear Sheet1!A2:D11 before a re-run (or point RANGE at a scratch tab while testing).

#9 Flag icons pad the country cells

Winner/Runners-up cells carry a stray non-breaking space next to the flag icon ("Uruguay " , " Argentina" — measured live). Text extraction normally trims whitespace; if a platform ever kept it, the Detour still passes (non-empty) and a one-time TRIM() in Sheets cleans column B/D.

#10 Why tr[{{index}}+1] is safe to template

After substitution the predicate is plain XPath arithmetic — tr[1+1] evaluates to tr[2] (confirmed in a live browser). Two equally valid alternatives, noted for completeness: give the Loop a list [2...11] so index is the row number directly and the +1 disappears; or use the self-documenting selector tr[position()>1][{{index}}] (“skip the header, take the i-th data row”). The count-based +1 form is chosen because it matches the only loop behaviour the library documents (index = 1...N).

#11 Parse number’s ambiguous output type

The library row contradicts itself: the description says “converts ... into its Int data type,” but the Output column reads “String.” The workflow doesn’t hinge on the resolution — the step earns its place as the row-validity gate (a non-numeric year errors and the loop skips that index). If the output is genuinely a string, year still appends correctly; column A’s numeric-vs-text nicety is the only thing at stake, and a one-time =VALUE() /format fix in Sheets settles it.

#12 Write authorization — why OAuth, not an API key

values.append is a write, so an API key alone is rejected (403). The chart uses Postman’s OAuth 2.0 Authorization Code flow (panel D); a Google service account with the sheet shared to its address is the equivalent server-to-server alternative. Either yields the Bearer token the Call API step needs.